Semiconductor Basics – Week 1

Fire EV Internship | CMOS & VLSI

During this week, I learned the basics of semiconductors and how they are used in electronic devices. Silicon is one of the most commonly used semiconductor materials. Its electrical properties can be changed by adding impurities through a process called doping.

I learned about electrons and holes, which are responsible for carrying current in semiconductors. Based on doping, semiconductors are classified into N-type and P-type. In N-type semiconductors, electrons are the majority charge carriers, while in P-type semiconductors, holes are the majority charge carriers.

I also studied PN junctions and how they work in a diode. When P-type and N-type materials are joined, a depletion region is formed near the junction. In forward bias, the diode allows current to flow more easily, whereas in reverse bias, only a small leakage current normally flows. I also learned how the voltage and current of a diode are related through its I–V characteristics.

Another important concept I learned was the effect of temperature on diode behaviour. As temperature increases, a typical silicon diode requires less forward voltage to maintain the same current. This helped me understand why temperature is an important factor when working with semiconductor devices.

These concepts are related to Battery Management Systems (BMS), where voltage and temperature monitoring are important for battery safety. Understanding semiconductor devices will help me learn how sensing and protection circuits work in a BMS.

Along with the theory, I practised circuit simulation using ngspice and plotted the simulation results using Python and Matplotlib. Overall, this week helped me understand the basics of semiconductor devices and gave me a starting point for learning MOSFETs, CMOS circuits, and their applications in battery management systems.

4. Checkpoint Questions and Answers

Q1. What is the difference between n-type and p-type silicon?

N-type silicon is doped with donor impurities that provide additional free electrons, making electrons the majority charge carriers. P-type silicon is doped with acceptor impurities that create holes, making holes the majority charge carriers.

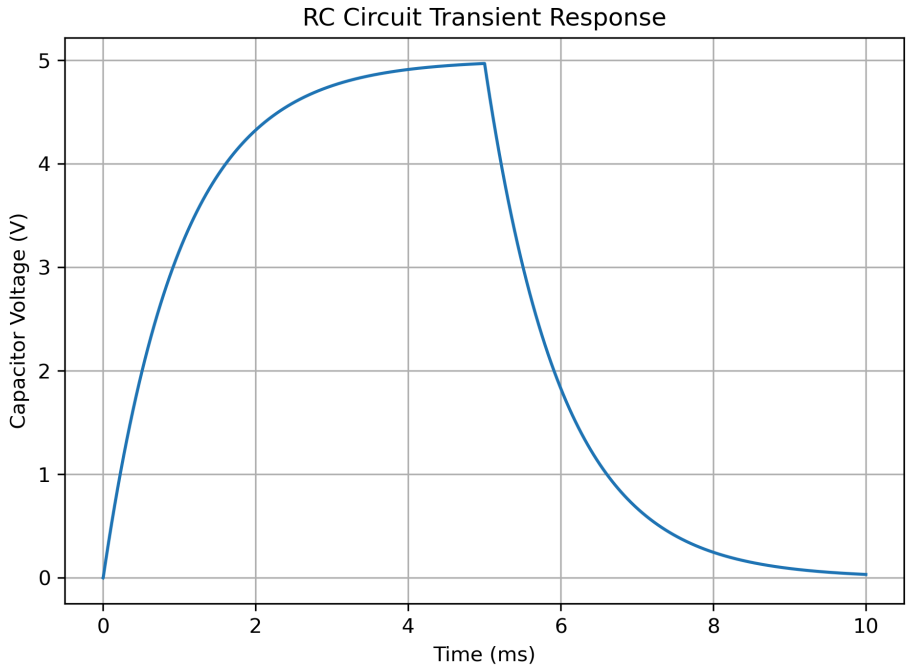
Q2. Why does a reverse-biased diode conduct almost no current?

Reverse bias widens the depletion region and increases the potential barrier, preventing majority carriers from crossing the junction easily. A small reverse leakage current can still flow due to minority carriers.

Q3. Why does a BMS need to measure cell voltage precisely?

Precise cell-voltage measurement helps the BMS monitor battery state, detect overvoltage and undervoltage conditions, support cell balancing, and protect lithium-ion cells from unsafe operating conditions.

RC-circuit Transistent Plot



Diode I–V comparison graph at 25°C and 75°C

